

in sleep mode constantly except when a connection is initiated. The actual connection times are only a few milliseconds (ms), unlike Bluetooth, which would take ~ 100ms.

Table 1 represents the Bill of Materials. Table 2 highlights the major differences between classic Bluetooth and BLE technologies.

BLE can communicate with other devices in two ways: broadcasting or connections. The broadcasting method by MAC address is used here. As per the guidelines established by the generic access profile (GAP), broadcast communication involves a BLE device (called a broadcaster) sending packets to any scanning device within its communication range. Any Bluetooth-compatible device (called the observer) can easily pick up the data broadcasted by the BLE device. This communication is unidirectional.

Broadcasting takes advantage of the advertising features of BLE to send a small packet (typically 47 bytes with 31 bytes customisable payload) using frequency hopping technique. BLE operates in the licence-free 2.4GHz ISM band, divided into 40 communication channels, having a bandwidth of 2MHz each. These 40 channels include 3 advertising channels and 37 connection channels.

TABLE 1
BILL OF MATERIALS

Components	Quantity
ESP32 (MOD1)	1
9V battery	1
BT66 (IC1)	1
BC547 (IC2)	1
Resistor 1kΩ (R1)	1
Smart watch	1
Small speaker	1
Jumper wires	10

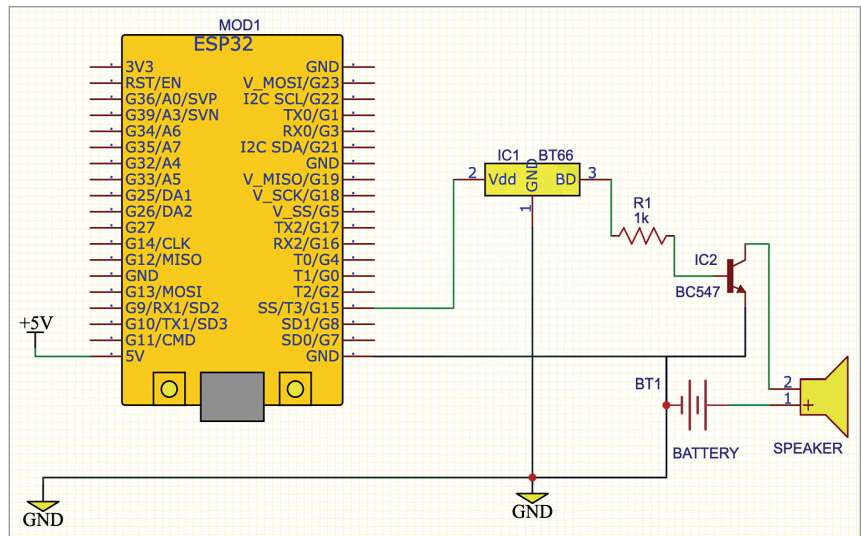


Fig. 3: Circuit diagram of BLE-based nearby person notifier

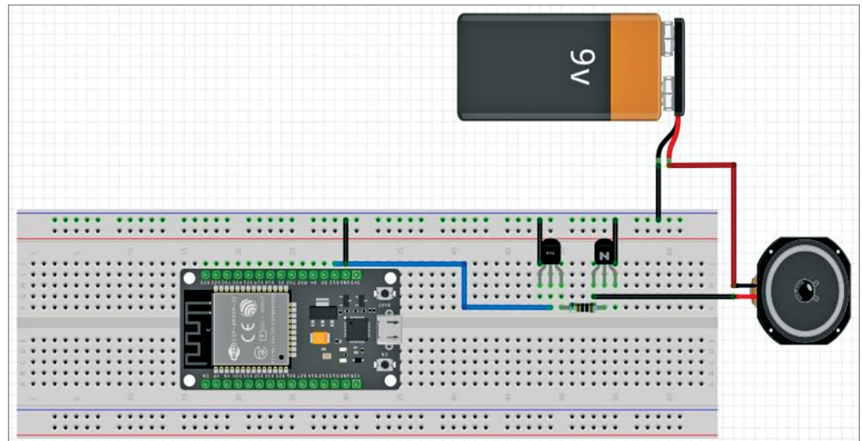


Fig. 4: Wiring of the proposed setup on a breadboard

TABLE 2
MAJOR DIFFERENCES BETWEEN CLASSIC BLUETOOTH AND BLE

Specifications	Classic Bluetooth	Bluetooth Low Energy (BLE)
Range	100m	Greater than 100m
Data Rate	1-3Mbps	1Mbps
Application Throughput	0.7-2.1Mbps	0.27Mbps
Frequency	2.4GHz	2.4GHz
Security	56/128-bit	128-bit AES with counter mode CBC-MAC
Robustness	Adaptive fast frequency hopping, FEC, fast ASK	24-bit CRC, 32-bit message integrity check
Latency	100ms	6ms
Time Lag	100ms	3ms
Voice Capable	Yes	No
Network Topology	Star	Star
Power Consumption	1W	0.01 to 0.5W
Peak Current Consumption	Less than 30mA	Less than 15mA